

# LLM Self-Recognition: Steering and Retrieving Activation Signatures

Thibaud Ardoin<sup>1</sup> Jonas Schäfer<sup>1</sup> Gerhard Wunder<sup>1</sup>

## Abstract

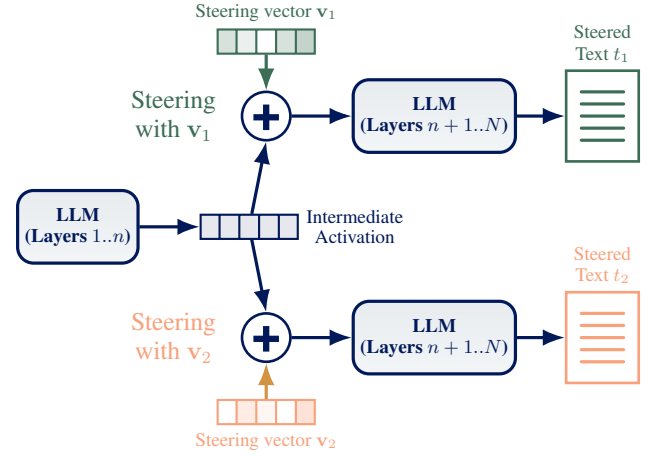
Recent advances in interpretability suggest that large language models (LLMs) implicitly encode signals in their generated text that enable self-recognition of their outputs. We demonstrate that this capability is reliable, even in low-entropy scenarios, and that it can be amplified through targeted intervention. By steering the internal residual stream during generation with a random sparse vector, we create a detectable fingerprint that enables attribution of a given text to a specific LLM. This signal is recoverable from the activations of an LLM used as a detector, achieving over 98% accuracy across multiple detection settings while preserving the quality of generated text. As AI-generated content proliferates, this approach offers a practical alternative to traditional detectors by leveraging the model’s natural representation structure for attribution rather than embedding a signal externally. Our contributions include: (i) establishing reliable self-recognition capabilities in LLMs, (ii) a simple steering mechanism enabling multi-LLM identification with no quality degradation, (iii) demonstrating that activation spaces contain exploitable structure for encoding signals without semantic interference.

## 1. Introduction

As large language models (LLMs) become increasingly deployed for content generation, concerns about the authenticity and traceability of AI-generated text have intensified. Unlike human-authored text, LLM generations often lack an intrinsic link to a responsible author, raising risks of knowledge base contamination and accountability gaps in

<sup>1</sup>Department of Computer Science, Freie Universität Berlin, Germany. Correspondence to: Thibaud Ardoin <thibaud.ardoin@fu-berlin.de>.

(a) Steering at generation to create a unique signature



(b) Extracting activations from the text and retrieving the steering vector

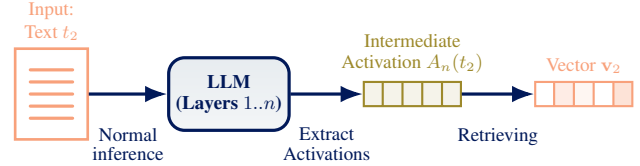


Figure 1. Steering and retrieval of LLM signatures. (a) During generation, steering vectors  $\mathbf{v}_1$  and  $\mathbf{v}_2$  are added to intermediate activations at layer  $n$ , creating model-specific signatures in outputs  $t_1$  and  $t_2$ . (b) To verify authorship, the generated text is passed through the same model to collect the activations at layer  $n$ . The source steering vector is retrieved via cosine similarity or a trained MLP classifier.

high-stakes domains. Distinguishing not only whether content is AI-generated, but which specific model produced it, is essential for auditing, attribution, and misuse prevention (Bjelobaba et al., 2025; Yousaf, 2025).

Current approaches to AI-generated text detection (AI-GTD) predominantly rely on watermarking methods that either modify token probability distributions (Kirchenbauer et al., 2023) or strategically select tokens to encode hidden information (Hou et al., 2023; Christ et al., 2024). Integrating these methods into the generation process incurs additional

overhead, and potential trade-offs between detection robustness and output quality have limited their widespread adoption (Sadasivan et al., 2023). Alternatively, AI-GTD can be approached post-generation using classifiers that exploit statistical properties of generated text (Mitchell et al., 2023) or classifiers trained on large labeled corpora. However, these detection methods are sensitive to distribution shifts across domains and do not naturally extend to distinguishing among multiple LLMs.

While our approach targets a different set of trade-offs than adversarially robust watermarking, it offers a fundamentally different lens on model attribution. Rather than relying on external watermarking mechanisms applied at the output token level, we investigate whether a signature can be embedded and recovered by exploiting the structure of the model’s internal representations. Specifically, we ask whether models naturally encode a fingerprint, how such signals can be detected, and whether an additional, easily recoverable signature can be embedded alongside them.

Recent work in interpretability has revealed that LLMs can indeed recognize their own generated outputs with non-trivial accuracy (Ackerman & Panickssery, 2025; Bowman et al., 2024). This capability, termed *self-recognition*, suggests that models implicitly encode model-specific information in their generations. At the same time, activation engineering has demonstrated that targeted interventions in internal activations can steer LLM behavior while largely preserving output quality (Panickssery et al., 2023; Liu et al., 2023). Together, these findings suggest that internal activations offer a principled mechanism for detecting and inducing model-specific generation signatures.

Building on these advances, we propose to use the internal representations of an LLM as the space in which authorship signals are detected. Furthermore, we show that inference-time steering can be used to deliberately inject distinct and recoverable signatures, enabling more reliable detection and supporting multi-model attribution without sacrificing generation quality. Our approach enables a flexible detection framework, covering a range of scenarios, from post-generation AI-GTD to the identification of multiple versions of the same model through steering. We release code and data to reproduce our experiments.<sup>1</sup> Our contributions in this work are threefold:

1. We empirically demonstrate *self-recognition*: even with very short texts in relatively low-entropy scenarios, LLMs can reliably distinguish their own output from human-written content.
2. We introduce a steering-based watermarking technique that injects a distinct, recoverable signature, enabling

accurate attribution across multiple identical LLM instances steered in different directions, without compromising generation quality.

3. We offer an analysis that sheds light on the capacity of internal representations to encode and retrieve a random signal.

## 2. Methods

### 2.1. Problem Setup

We consider *white-box* detection and attribution of LLM-generated text using internal activations of a target model  $M$ . We distinguish two setups:

- (i) **AI-generated text self-recognition**: Given a text  $t$ , decide whether it was generated by  $M$  or written by a human.
- (ii) **Multi-model attribution**: Decide which *steered* variant of the same base model  $M$  produced a given text  $t$ .

**Threat model**: At detection time, we assume white-box access to the internal activations of  $M$  when processing a given text  $t$ . We consider both a *prompt-conditioned* setting, in which the prompt is available to the detector, and a *prompt-agnostic* setting, in which only the generated text  $t$  is observed.

### 2.2. Activation Extraction

Given a model  $M$  with  $L$  layers, let  $l \in \{1, \dots, L\}$  denote a layer and  $\mathbf{x} = [x_0, x_1, \dots, x_{n-1}]$  denote the token sequence used for activation extraction (prompt + completion, or completion only in the prompt-agnostic setting). Let  $\mathbf{A}_l(x_i) \in \mathbb{R}^d$  be the intermediate activation at layer  $l$  downstream from token  $x_i$ . By extension, for a text  $t$ , we describe the activation sequence as  $\mathbf{A}_l(t) \in \mathbb{R}^{n \times d}$ . We extract  $\mathbf{A}_l(\cdot)$  at a fixed, consistent location within each transformer block. The extraction layer  $l$  is optimized independently for each model and corresponds to an intermediate layer near the middle of the network.

### 2.3. Steered LLM Generation

To embed a reliably detectable signature in the generated text, we steer the model during generation. Given a directive vector  $\mathbf{v} \in \mathbb{R}^d$  and an amplification coefficient  $\alpha > 0$ , we add the scaled vector to the intermediate activation at layer  $l$  at each token generation step:

$$\mathbf{A}_l(x_i) \leftarrow \mathbf{A}_l(x_i) + \alpha \mathbf{v}.$$

Intuitively, this intervention nudges the internal representation trajectory in a consistent direction, leading to a text  $t_v$  influenced by  $\mathbf{v}$ . At detection time, this influence can be retrieved in the activations  $\mathbf{A}_l(t_v)$  of  $M$ .

<sup>1</sup><https://github.com/Thibaud-Ardoin/LLM-Self-Recognition.git>

For simplicity, steering and activation extraction use the same layer  $l$ . The steering direction  $v$  is chosen randomly to create a unique watermark in the multi-model attribution setting. While prior work has largely focused on dense interventions that modify all hidden dimensions (Li et al., 2023), recent studies suggest that sparse steering vectors induce fewer undesired perturbations and lead to more stable model behavior. Such sparsity can be achieved either by leveraging sparse autoencoders (Bayat et al., 2025) or by directly enforcing sparsity in the activation space (Ardoin et al., 2025). We adopt the latter approach and construct steering vectors with only a small fraction of non-zero dimensions. We empirically evaluate this design choice in Section 3.4.

## 2.4. Attribution

To highlight the intrinsic self-recognition capability of LLMs, we adopt a minimal linear probe operating on averaged internal activations. Multi-model attribution, however, poses a more challenging classification problem and therefore needs a more expressive probing approach.

### 2.4.1. LINEAR CLASSIFICATION FOR SELF-RECOGNITION

To obtain a fixed-dimensional representation of a text  $t$  that is independent of its token length  $n$ , we aggregate the token-level activations extracted in Section 2.2 by simple averaging:

$$\mathbf{r} = \frac{1}{n} \sum_{i=0}^{n-1} \mathbf{A}_l(x_i) \in \mathbb{R}^d \quad (1)$$

We use a lightweight linear discriminant analysis (LDA) classifier to distinguish *human* from *LLM-generated* text based on the extracted, standardized representations  $\mathbf{r}$ . LDA gives an affine decision function, which we use to score texts and classify them using a threshold  $\tau$  selected to satisfy a target false-positive rate. In high-dimensional regimes, the empirical covariance estimate of the LDA may be ill-conditioned. We therefore regularize the covariance with the Ledoit–Wolf shrinkage estimator (Ledoit & Wolf, 2004). In this setting, we use an 80/20 train/test split.

### 2.4.2. MULTI-MODEL ATTRIBUTION

To learn to distinguish among  $K$  different steering vectors  $(\mathbf{v}_k)_{k \in [1, K]}$ , we construct a labeled dataset of activations. Let  $M_{\mathbf{v}_k}(p) = t_{\mathbf{v}_k, p}$  denote the text generated by model  $M$  steered with vector  $\mathbf{v}_k$  in response to prompt  $p \in \mathcal{P}$ :

$$\mathcal{D} = \{(\mathbf{A}_l(t_{\mathbf{v}_k, p}), k) \mid k \in \{1, \dots, K\}, p \in \mathcal{P}\}.$$

We divide the dataset  $\mathcal{D}$  with a 70/10/20 train/validation/test split by partitioning the prompt set  $\mathcal{P}$ , ensuring no prompt

overlap across splits. We then train a multi-layer perceptron (MLP), which provides greater expressive capacity than LDA. The MLP has two hidden layers of width 32, trained for a single epoch to predict the steering index  $k$  of a given extracted activation. As detection operates at the *token level*, it enables attribution on arbitrary-length texts. However, tokens later in a sequence encode broader context due to causal attention over preceding tokens. To obtain *text-level* predictions, we aggregate token-level classifications by majority voting across the entire generated sequence.

## 3. Experiments

Our experiments aim to assess the effectiveness of using internal activation patterns for text authorship classification and to validate the proposed steering-based watermarking method. We primarily conduct our experiments using Llama-3.1-8B, chosen for its wide availability and favorable performance-to-size trade-off. To assess the generality of our approach across model families, we additionally evaluate Mistral-3-8B, a state-of-the-art mixture-of-experts (MoE) model in the 8B-parameter range. To study scalability, we further include experiments on the smaller Llama-3.2-1B and Llama-3.2-3B models. For question-answering benchmarks, we use the instruction-tuned variants of all models.

To evaluate robustness across a range of generation regimes, we employ diverse datasets ranging from low-entropy constrained outputs to high-entropy open-ended completions. For low-entropy tasks designed to demonstrate the self-recognition capability of LLMs, we use the XL-Sum summarization dataset (Hasan et al., 2021). We restrict our experiments to the English subset of the human-authored BBC news articles and summaries. For long-form question answering, we use the ELI5 dataset (Guo et al., 2023), consisting of open-ended questions extracted from the *Explain Like I’m Five* subreddit. Finally, to evaluate performance under higher-entropy generation, we curate a *Fresh News* dataset from news articles published after the training cutoff of the evaluated models.

### 3.1. Self-Recognition of Unaltered LLM-Generated Text

To evaluate the capability of LLMs to distinguish texts they generated themselves from human-authored ones, we prompt the LLM to generate short summaries of given news articles and compare them against the corresponding human-authored summaries. This setting enables tight control of confounds, particularly regarding text length. Furthermore, summarizing news articles in only 1–2 sentences is inherently low-entropy, as the summary must cover the article’s key information, leaving little stylistic freedom. Authorship attribution in this setting is therefore more challenging